MLR780 Air Conditioning and Refrigeration 2021

Assignment 6
Cooling Towers

Instructions

- This individual assignment is to be submitted online before midnight on **06 June 2021**.
- Repetitive calculation will be needed. Use a programming platform or a spread sheet of your choice. Use CoolProp for thermodynamic properties.
- Prepare a technical report that supplies all relevant equations, methods and input values, and results. Include everything that is to be graded in your report. Name it: "MLR780-2021-Assignment6-SURNAME.pdf"
- Each submission (PDF) **must** be accompanied by an official cover page obtainable from Click-UP, **as well as** your supporting documentation to prove you performed the analysis.
- The supporting documentation will not be graded, but will be checked: It must be a soft version of the script (a *.txt text file) or spread sheet with the following naming convention: "MLR780-2021-Assignment6-SURNAME.* " (The "*" represent the file extension, which will depend on the program you used – it should not be "pdf").
- **Only the PDF report will be graded.**
- Keep a time sheet documenting the time it took you to solve the questions and indicate this on the first page of your submission. Time spent on the lecture content / example / programming revision should not be included.

Outcomes:

This assignment is designed to assist you develop a basic understanding of the behaviour a cooling tower and how to perform first order modelling there-of based on information obtained from a manufacturer catalogue. The methods dealt with here should only be used when catalogue entries are not detailed enough to cover the conditions you are interested in. When accepted measuring standards are employed, experimentally obtained performance data are more accurate than results obtained from modelling.

Background:

Consider the information supplied in the Evapco Catalogue for Cooling Tower Model LPT326. (Download this catalogue from Click-UP.) Assume that the information in the catalogue is valid for an atmospheric pressure of 101.325 kPa (altitude, z = 0 m) and 50% relative humidity.

1. The cooling tower is employed to deliver cooling to a water-cooled condenser by supplying it with water at 26°C. Identify the matching catalogue entry for a range of 10°C and an approach of 7°C. List the entering water temperature, inlet wet-bulb temperature, the water volume flow rate, and the air volume flow rate.

Use 4 horizontal divisions and 3 vertical division, resulting in 12 equal area divisions in total (n = 1 to 12) to analyse the thermal states within the cooling tower. Refer to Figure 1 showing the internal division numbers. Assume that the $h_c A/c_{pm}$ value is uniform in all the sections.

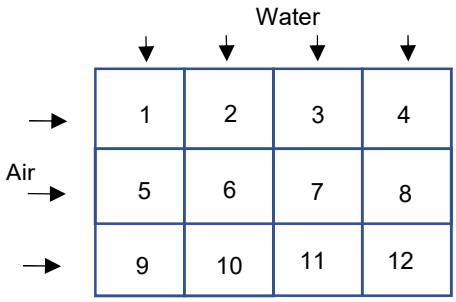


Figure 1 Equal area internal divisions

2. For the answer from Question 1: Based on an assumed $h_c A/c_{pm}$ value of 5 kW/kJ/kg for this entry of question 1 (not yet the correct value), calculate the associated **water outlet temperature** of the cooling tower. Show all equations and input values and tabulate your converged calculated values for each division as indicated in Table A.

Table A

n	$t_{w,in}$	$h_{a,in}$	$h_{i,ave}$	$t_{w,out}$	$h_{a,out}$	$\dot{Q}_n$
1						
2						
…						
12						

3. Repeat question 2 for several other $h_c A/c_{pm}$ values to determine the impact of the NTU on the tower's exit water temperature and the total heat transfer rate. Tabulate your results as in Table B, and plot them. Comment on and discuss the tends you observe. Explain the meaning of the trends. Tabulated results in the format of Table A is not needed in this question.

Table B

NTU	$t_{w,out}$	$\dot{Q}_{TOT}$

4. State and motivate the correct $h_c A/c_{pm}$ value for the case in Question 1.
5. Identify another (singular entry) in the catalogue with which you can validate your answer in Question 4. Repeat Question 1 for that entry. Present your converged calculations using Table A. Discuss your validation and its meaning.

Higher difficulty question

6. Use the remaining available data for Model LPT326 to check if the adjustment technique on $h_c A/c_{pm}$ in terms of the water flow rate as described on slide 07-23 of the lecture notes are also valid for cross flow coiling towers. Present relevant details, tabulated findings and suitable plots to support your conclusion.

INDICATE HOW MUCH TIME YOU SPENT ON THIS ASSIGNMENT BY PUTTING IT AT THE START OF YOUR REPORT.